

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

① Given $2A+B = \begin{bmatrix} 0 & -1 \\ 9 & -5 \end{bmatrix}$, $A-B = \begin{bmatrix} 6 & -2 \\ 0 & -1 \end{bmatrix}$

then = (i) $A+B = ?$

(ii) Find the matrix C , satisfied $B+C=2A$

② For what values of h , y will not be in $\text{span}\{v_1, v_2, v_3\}$ if

$$v_1 = \begin{bmatrix} 1 & -1 & -2 \end{bmatrix}^T, v_2 = \begin{bmatrix} 5 & -4 & -7 \end{bmatrix}^T, v_3 = \begin{bmatrix} -3 & 1 & 0 \end{bmatrix}^T$$
$$y = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$$

③ For any vector $v = (v_1, v_2)$ in $\mathbb{R}^2$, define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

by $T(v_1, v_2) = (v_2 - v_1, v_1 + v_2)$. Show that T is a linear transformation

④ let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Give a geometric description of the transformation $x \mapsto Ax$

⑤ Find the eigenvalues and eigen vectors of the matrix

$$\begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix}$$